

DAY 01 · LAB

Size a Facility.

DURATION

90 minutes

TEAMS OF

Three

YOU NEED

Pen, paper, a calculator

SUMS INVOLVED

Ten, none of them hard

This morning you looked inside a data centre.
This afternoon **you build one.**

The whole class designs the same facility, so you can compare answers. Then each team gets one curveball that nobody else has.

By the end you will understand why this industry describes buildings in **kilowatts** rather than square metres, and why the biggest cheque an operator writes each year goes to the electricity company.

THE ONE IDEA BEHIND ALL OF IT

Everything a server draws from the wall comes back out as heat.

A room using 200 kW of electricity is a 200 kW heater, and something has to remove all of it, which takes more electricity again.

The best buildings in the world waste about a tenth on top. Ordinary ones waste half as much again. That gap is worth millions.

THE SIX NUMBERS • COPY THESE ONTO YOUR PAGE

	RULE	NUMBER
1	Each ordinary server draws about	0.5 kW
2	Add for switches and storage	10% on top
3	A rack holds about	20 servers
4	Air cooling stops working above	15 kW per rack
5	For every 1 kW of computing, the building uses	1.5 kW
6	Each rack needs floor space of about	3 m²

THREE MORE TO LOOK UP

Hours in a year	8,760
Electricity	USD 0.10 per kWh
Cost to build	USD 10M per MW

Rule 5 has a name. It is called PUE, and it is the single number the industry uses to judge whether a building is any good.

———— DONE TOGETHER ON THE BOARD • 100 SERVERS, 400 W EACH, 25 TO A RACK

STEP		THE SUM	ANSWER
1	Power for the servers	100×0.4	40 kW
2	Add 10% for the rest	$40 + 4$	44 kW of computing
3	Racks needed	$100 \div 25$	4 racks
4	Power per rack	$44 \div 4$	11 kW, under 15
5	Power for the building	44×1.5	66 kW
6	Heat to remove	same as step 2	44 kW
7	Floor space	4×3 , doubled	24 m²
8	Electricity per year	$66 \times 8,760$	578,000 kWh
9	Annual bill	$\times \text{USD } 0.10$	USD 58,000 a year
10	Cost to build	$0.044 \text{ MW} \times 10\text{M}$	USD 440,000 once

LOOK AT THE LAST TWO LINES

The building is paid for once. The electricity is paid for every year, forever, before anybody is employed and before a single server is replaced.

That relationship is the entire economics of this industry.

————

SAME TEN STEPS • BIGGER BUILDING

Your university is building a research computing facility.

Servers	400
Each draws	0.5 kW
Per rack	20
PUE	1.5
Electricity	USD 0.10

QUESTION A • HEAVIER SERVERS

At step 4 you found the power per rack. Is it under 15 kW?

What would you have to change if your servers drew **1 kW each** instead of 0.5 kW, and you still wanted 20 in a rack?

QUESTION B • RELIABILITY

Choose a level from the table, then write **one sentence** saying why it is right for a university research facility.

LEVEL	DOWNTIME PER YEAR
Basic One power supply, one cooling system	about 29 h
Better Spare parts, still one path in	about 22 h
Good Two of everything, service without stopping	about 1.6 h
Best Two live systems, survives a fault mid service	about 26 min

Each level up costs more to build and more to run, because you are powering equipment that sits idle waiting for a bad day.

Does your answer look right?

Check before you go on to your curveball. If any of these is far off, find the sum that went wrong rather than carrying the error forward.

YOUR NUMBER	SHOULD BE ROUGHLY
Computing power	200 to 250 kW
Racks	Around 20
Power per rack	Around 10 kW
Building power	300 to 350 kW
Floor space	100 to 150 m²
Annual bill	USD 250,000 to 300,000
Cost to build	USD 2 million to 3 million

The two places people slip: adding the 10% to the wrong number, and multiplying the electricity price by the computing load instead of the building load.

YOUR TEAM DRAWS ONE CARD • NOBODY ELSE HAS YOURS

CARD A • THE AI GROUP HAS BEEN FUNDED

They want to add **20 GPU servers**. Each draws **10 kW**, and only **4 fit in a rack**.

- How much power do the GPU servers add on their own?
- What is the new total for computing, and for the building?
- What is the power per rack in the GPU racks?
- Compare that to rule 4. Can the room you designed cool them?

Tell the class in one sentence what has to change, and roughly what it does to the annual bill.

CARD B • THE EFFICIENCY QUESTION

An architect says she can deliver the same facility at **PUE 1.2** instead of 1.5, for an extra **USD 400,000** to build.

- Recalculate the building power and the annual bill at 1.2
- How much does she save you each year?
- Over ten years, how much is that?
- Is the extra USD 400,000 worth spending?

Tell the class the year in which the better building has paid for itself.

CARD C · THE POWER COMPANY SAYS NO

- The campus substation can supply the building with **250 kW** in total, and an upgrade takes two years.
- Your design needs more than that. By how much are you over?
 - Working backwards, how many servers actually fit inside 250 kW?
 - Now try it again assuming you build the efficient version at PUE 1.2. How many servers fit then?

Tell the class how many servers good engineering just bought you, without any help from the power company.

CARD D · THE SUSTAINABILITY QUESTION

- Every kilowatt hour on this grid produces about **0.5 kg of CO2**.
- How many tonnes does your facility produce each year? (1,000 kg = 1 tonne)
 - An average petrol car produces roughly 4 tonnes a year. How many cars is your building?
 - If you moved this facility to a cleaner grid, at 0.05 kg per kWh, what happens to the number?

Tell the class one non-technical reason a university might build here anyway.

ONE CARD • WRITE IT BIG ENOUGH TO HOLD UP

Servers	COUNT
Racks	COUNT
Power per rack	KW • AIR COOLED? Y / N
Computing power	KW
Building power	KW
Floor space	M ²
Annual electricity bill	USD
Cost to build	USD
Reliability level, and why	ONE SENTENCE
Our curveball card, and the answer	ONE SENTENCE

WHEN YOU PRESENT

Ninety seconds. Lead with the **building power in kilowatts**.

That is the number the industry leads with, and by tomorrow you will know why.

WHAT TO TAKE AWAY

A data centre is a power station with **computers** attached.

THE REAL LIMIT

Space is rarely the constraint.
Power almost always is

THE REAL COST

The building arrives once. The
bill arrives every year

WHY EFFICIENCY

It decides whether you fit inside
your supply at all

HONESTY

Every number today is a rule of
thumb. Real projects argue for
months